HEIRSTONE CONSULTING

TRADE CORRIDORS | AFRICA

Where cargo stops

Ports, dry ports, border towns and junctions after the African Continental Free Trade Area, and where warehousing and industrial land should follow



THE REPORT IN SIX NUMBERS

106.7 h — average dwell of containerised cargo at the port of Mombasa in 2025, more than twice the port's 48-hour target

1.8% — share of its 450,000 TEU design capacity that Kenya's Naivasha inland container depot used in 2025; Nairobi's depot used 69%

21% — share of AfCFTA country pairs in which both sides had gazetted their tariff schedules by April 2025 (Heirstone analysis)

+161% — population growth of Chirundu, on Zambia's first one-stop border post, between the 2010 and 2022 censuses; Lusaka grew 27%

3rd — Port Said's world rank in the World Bank's 2024 port index, with Tanger Med 5th; most Sub-Saharan gateways ranked 309th to 405th in 2023

US$4-7 — prime logistics rent per m² a month in seven corridor cities in H2 2025, with yields from 8% near the coast to 13% in Kampala

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ABOUT THIS REPORT

Heirstone Consulting advises corporates, investors, developers and governments across the Middle East and Africa on markets, feasibility and the best use of land. This report draws on public sources current in 2026: port authorities and corridor agencies, the World Bank, Afreximbank, national censuses, company and development finance disclosures, broker research and the specialist press. Every figure is referenced in the exhibit sources and in the notes.

Where we have combined published figures into a new one, such as capacity in use, hours saved, capital per tonne or growth against the capital city, the calculation is shown and labelled Heirstone analysis. Local-currency figures are converted at World Bank annual average rates for the year stated. Map positions are approximate town centres.

WHAT WE FOUND

The free trade area is likely to raise land values where cargo has to stop and where the hours it loses are being cut. Much of the new line capacity will wait for traffic.

Corridors are usually sold by the kilometre. The evidence in this report points to the nodes: much of a container's time between ship and customer is lost at the port gate and in the inland clearance yard, and the land that gains value is the land that removes those hours.

1 **The treaty has run ahead of the trade.**

Fifty countries have ratified the AfCFTA, but only 23 had gazetted their tariff schedules by April 2025, so on that count about 21% of country pairs could use its tariffs. Intra-African trade was 14.7% of Africa's trade in 2024, against 21.2% inside ASEAN. The World Bank attributes US$292bn of the agreement's projected US$450bn income gain to trade facilitation.

2 **The port gate now costs more hours than the border.**

Containerised cargo waited an average of 106.7 hours at Mombasa in 2025, more than four days and over twice the port's 48-hour target, before any journey inland began. Trucks need 30 to 35 hours to move a box from Dar es Salaam port to the Kwala dry port, against about an hour by rail. Land borders with one-stop posts take about 40 minutes to 12 hours.

3 **New capacity is a poor guide to cargo.**

Naivasha's depot used 1.8% of its capacity in 2025 and the Addis Ababa-Djibouti railway 16 to 32%, while Nairobi's depot ran at 69%. The Ethiopian line's capital cost was US$1,100 to 1,300 per tonne carried in 2025, and Lobito's 4.6 Mt target is 1.4 times the Benguela line's 1973 peak.

4 **Africa's ports have split into two economies.**

Port Said ranked 3rd and Tanger Med 5th in the World Bank's 2024 port index; most Sub-Saharan gateways ranked 309th to 405th in the 2023 edition. At the hubs land value sits in free zones, at the gateways in land that relieves the gate.

5 **Border towns grow where cargo breaks bulk.**

Between Zambia's 2010 and 2022 censuses Chirundu grew 161% and Nakonde 118%, against 27% for Lusaka; Beitbridge is five times its 1992 size. Chirundu's one-stop gains eroded as traffic grew.

6 **Rents are flat in dollars; land carries the upside.**

Prime logistics rents sit at US$4 to 7 per m^2 a month, and only Lagos and Kampala are above their 2021 level. Yields of 12 to 13% in Kampala and Lusaka, against 8 to 10% near the coast, show where faster corridors could add most value, and serviced land at working nodes is full.

WHAT IT MEANS

Corporates should hold regional stock where cargo clears customs and trains run, and check each trade lane against the list of gazetted schedules before counting on tariff savings.

Investors and lenders should underwrite measured throughput and hours saved, treat design capacity as a ceiling, and expect most of the return from land and yield compression.

Governments and port authorities should publish dwell times and release serviced land beside working gates and depots; in the cases we reviewed, designation alone has not brought occupiers.

34

nodes in 22 countries in the Heirstone Corridor Node Benchmark

16

nodes scored on the five-part Heirstone Node Test, from 4 to 8 points out of 10

CHAPTER 1

01

The treaty runs ahead of the trade

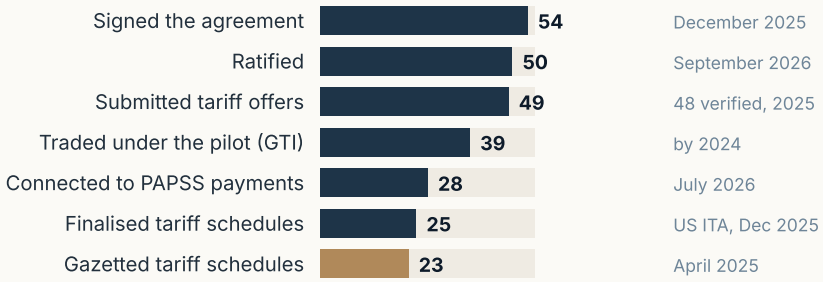
The African Continental Free Trade Area is close to universal on paper. Fifty-four countries have signed it and 50 had ratified by September 2026; Benin, Libya, South Sudan and Sudan have signed without ratifying.

In practice it is narrower. A preferential tariff applies only when both the exporting and the importing country have put their schedule of concessions into domestic law, and only for goods that meet agreed rules of origin.

EXHIBIT 1

Fifty countries have ratified the AfCFTA; 23 had put their tariff schedules into law

Number of AU member states at each stage of AfCFTA implementation, latest date available for each stage



Source: tralac, Status of AfCFTA ratification (4 Sep 2026) and 2025 Annual Conference two-pager; US International Trade Administration, Ghana AfCFTA update (Dec 2025); PAPSS (9 Jul 2026).

By April 2025, 23 State Parties had gazetted their provisional schedules, Nigeria being the 23rd. If each of the 23 can trade on AfCFTA terms with each of the others, about 21% of all ordered pairs of State Parties could use the agreement's tariffs on that count (Heirstone analysis: 23 x 22 over 50 x 49). The US International Trade Administration lists 25 countries with finalised schedules in December 2025, one of them South Sudan, which has not ratified; on the other 24 the share is about 23%. Côte d'Ivoire, Senegal, the DRC, Angola and Zimbabwe are not on that list.

Rules of origin are agreed for 92.43% of tariff lines. Textiles and clothing (HS chapters 60 to 62) and vehicles (chapter 87) remain open, so two of the sectors that fill large distribution warehouses elsewhere gain nothing yet. Payments are also partial: the PAPSS settlement system reached 28 countries when the central bank of the six CEMAC states joined in July 2026.

21%

of ordered country pairs where both sides had gazetted AfCFTA schedules, April 2025 (Heirstone analysis)

92.43%

of tariff lines with agreed rules of origin; clothing and vehicles are still outstanding

US$5,000

consignment value up to which an exporter can self-declare origin instead of obtaining a certificate

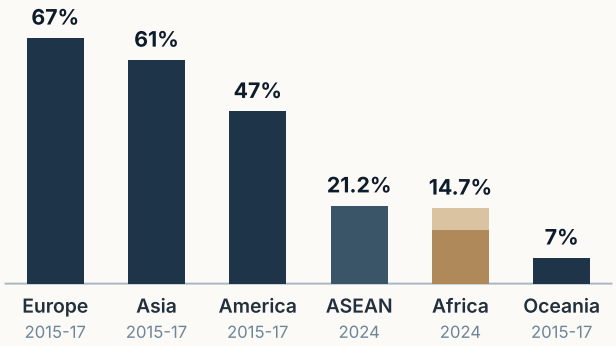
THE TRADE THAT EXISTS

Intra-African trade is small and concentrated on a handful of corridors

EXHIBIT 2

Africa's intra-regional trade share is about a fifth of Europe's and 70% of ASEAN's

Intra-regional trade as a share of each region's total trade, %. Gold extension: Africa with informal trade, on an 11 to 40% under-reporting estimate (Heirstone analysis)



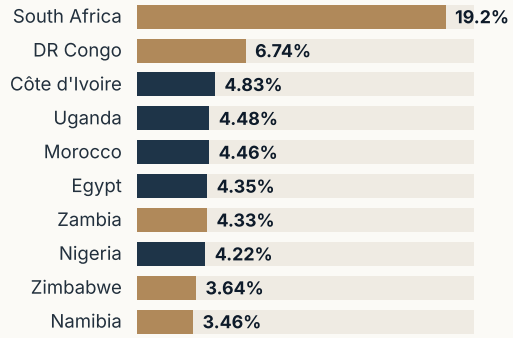
Source: Afreximbank, African Trade Report 2025 via Agence Ecofin (US$220.3bn of US$1.5tn, 2024; Heirstone calculation); UNCTAD, Economic Development in Africa Report 2019 (2015-17); 57th ASEAN Economic Ministers' report via Vietnam Ministry of Industry and Trade (2024); Brookings (2021) for informal trade.

Afreximbank put intra-African trade at US$213.8bn in 2025, on a series it revised between editions (2024 moved from US$220.3bn to US$202.7bn). On the earlier vintage it was 14.7% of Africa's US$1.5tn of merchandise trade in 2024. That is about a fifth of Europe's share, measured by UNCTAD at 67% in 2015 to 2017, and about 70% of ASEAN's 21.2% in 2024. Informal trade, which research cited by Brookings puts at 11 to 40% under-reported, would take Africa's share to roughly 16 to 21%.

EXHIBIT 3

Five economies on the North-South and Copperbelt system hold 37% of intra-African trade

Share of intra-African merchandise trade by country, 2025, %; gold marks the North-South, Copperbelt and Walvis Bay corridor system

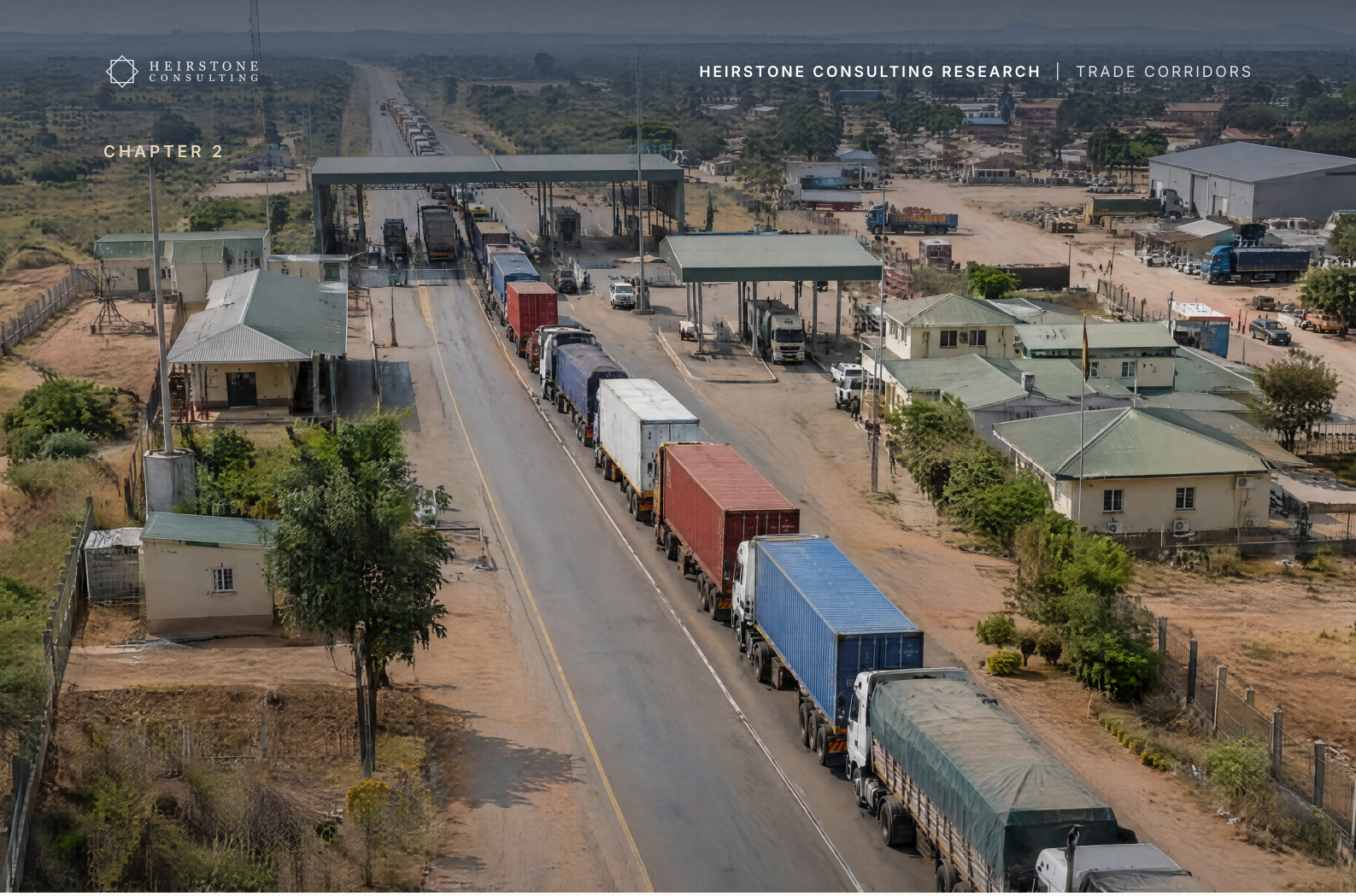


Source: Afreximbank, African Trade Report 2026, via Agence Ecofin (8 Jul 2026); corridor grouping and sum by Heirstone.

The trade that exists is concentrated, and so are the routes that carry it. South Africa, the DRC and Côte d'Ivoire account for 30.8% of intra-African trade and the top ten for 59.75%. Five economies on the North-South, Copperbelt and Walvis Bay system account for 37.4% between them (Heirstone analysis). South Africa exported US$31.1bn to the rest of Africa in 2025 and imported US$10.0bn, about a third as much, which suggests that trucks leaving South Africa run fuller than those returning.

> *The World Bank attributes about 65% of the AfCFTA's projected income gain to trade facilitation, the work that ports, depots and borders do.*

The World Bank estimates that the agreement could raise African incomes by US$450bn by 2035 and attributes US$292bn of that to trade facilitation, with the rest from tariff cuts. Much of that facilitation happens at ports, inland depots and border posts, so the rest of this report measures corridors in hours and tonnes.



02

Where the hours go

On the corridors that publish time data, a container can lose days at the port gate and in the clearance yard before its journey inland begins.

The Northern Corridor publishes the best time data on the continent. In 2025 containerised cargo spent an average of 106.7 hours in the port of Mombasa, more than double the target of 48 hours, according to the corridor authority's 21st Transport Observatory Report (August 2026). On that average, a box spent more than four days in the port before its journey inland to Nairobi, Kampala or Kigali began. The 58.7 hours above target alone exceed the largest saving in the one-stop border post surveys we compile (Exhibits 4 and 13).

The Central Corridor shows the same pattern at a smaller distance. Tanzania opened the Kwala dry port, 502 ha in Kibaha district, in July 2025. By road, a container needs 30 to 35 hours to get there from the port of Dar es Salaam, 24 to 30 of them spent on port entry and exit. By standard gauge railway the transfer takes 45 to 60 minutes.

Land borders, long the villain of African logistics, are now the smaller problem on the main corridors. Where one-stop border posts have opened, crossing times at Busia, Mutukula, Taveta-Holili, Mirama Hills and Rusumo fell by about 60 to 90%, to between about 40 minutes and 5 hours.

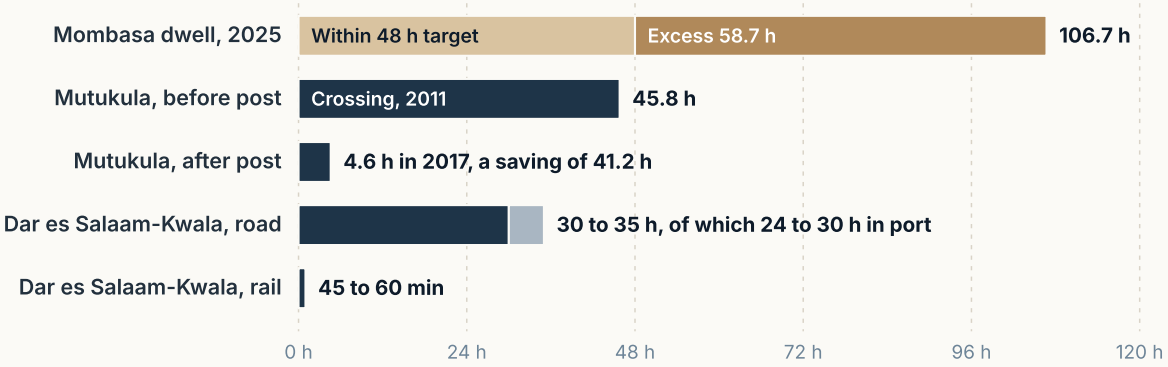
THE TIME BUDGET

At Mombasa, cargo spends more than four days in port, over twice the target

EXHIBIT 4

At Mombasa, the hours above the dwell target outweigh the largest surveyed border-post saving

Hours. Average containerised cargo dwell at Mombasa in 2025, split at the 48-hour target; Tanzania-to-Uganda truck crossing at Mutukula before and after its one-stop post, the largest saving in Exhibit 13; container transfer from Dar es Salaam port to the Kwala dry port in 2025, lighter bar showing the published range



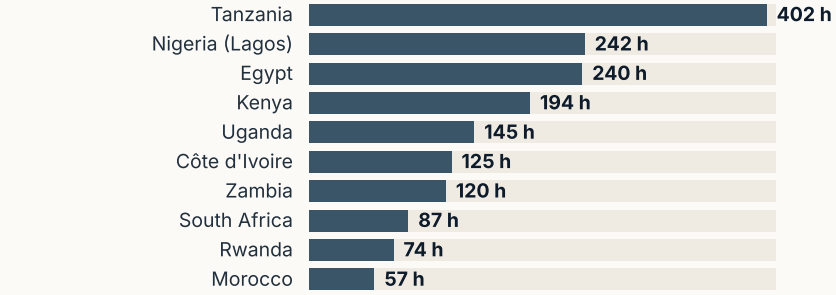
Source: Northern Corridor Transit and Transport Coordination Authority, 21st Transport Observatory Report (Aug 2026); TradeMark Africa OSBP portal (Mutukula); The Chanzo (Aug 2025, Kwala); excess and saving by Heirstone.

EXHIBIT 5

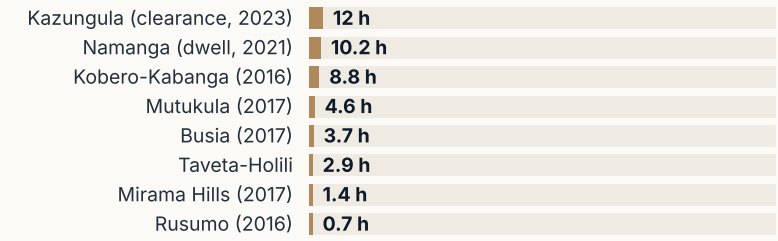
Import clearance at the port takes days; land borders with one-stop posts now take hours

Doing Business 2020 import border compliance (2019 data) at each country's main port; truck crossing, clearance or dwell time after a one-stop border post opened (latest survey). Measures differ and are not directly comparable

IMPORT CLEARANCE AT THE MAIN PORT, DB2020



LAND BORDER AFTER A ONE-STOP POST



Source: World Bank, Doing Business 2020 economy profiles; TradeMark Africa OSBP portal; Central Corridor TTFA (2016); JICA (2026); Freight News (Aug 2023, Kazungula).

The World Bank's last comparable measure of import clearance, Doing Business 2020, put border compliance for imports at 402 hours in Tanzania and 194 in Kenya, against 57 in Morocco. The series ended in 2021 and some ports have improved since: DP World reports cutting ro-ro discharge at its Dar es Salaam berths from over 300 hours to under 28. On the World Bank's more recent Logistics Performance Index (2023), Egypt scores 3.6 for timeliness and 3.1 overall, joint second in Africa.

A one-stop post saves between 1 and 41 hours a crossing on the surveys in Exhibit 13. The excess of Mombasa's dwell over its target is larger than any of those savings.

58.7 h

excess of Mombasa's 2025 dwell over its 48-hour target (Heirstone analysis)

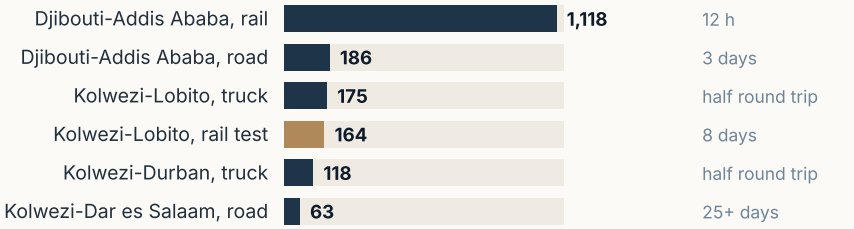
SPEED AND PRICE

Rail transforms a corridor only when it runs, and road prices have hardly moved in nominal terms

EXHIBIT 6

The Lobito rail test moved copper no faster per kilometre than a truck on the same route

Effective speed: straight-line distance between town centres divided by published door-to-door transit time, km a day; published time at right. Round-trip truck times are halved



Source: Heirstone analysis of Djibouti Ports and Free Zones Authority (Jan 2017, rail time as announced at opening), IRIS (Nov 2025, citing Ivanhoe Mines) and Freight News (2025); straight-line distances from GeoNames town centroids.

Exhibit 6 converts published transit times into straight-line kilometres per day, a crude but comparable measure of how fast cargo moves between two points. When rail runs, it changes the corridor: at its opening in 2017, Djibouti's port authority said the Addis Ababa-Djibouti railway would cut the journey from three days by road to twelve hours, about 1,100 straight-line km a day.

The Lobito test is a corrective. A first cargo of copper concentrate took eight days by rail from Kolwezi to Lobito in late 2023, against at least 25 days by road to Dar es Salaam, according to Ivanhoe Mines communications cited by IRIS. Per straight-line kilometre that is about 164 km a day, against about 175 km a day for half a truck round trip on the same route. Lobito's advantages are a single border crossing and a lender claim of up to 30% lower cost; Freight News reports truck round trips from Kolwezi of 14 to 16 days to Lobito against 35 to 40 to Durban.

Prices per kilometre have barely moved. The World Bank's 2009 corridor study put road freight prices on four African corridors at 6 to 11 US cents per tonne-km in 2007, against 5 cents in China (2007) and France (2009). Northern Corridor rates of US$1.8 to 2.4 per truck-km in 2025 work out at roughly 6.4 to 8.6 cents per tonne-km at an assumed 28-tonne payload (Heirstone analysis), close to Mombasa-Kampala's 8 cents in 2007 in nominal terms.

35%+

of Mombasa's containers moved empty in 2025

US$0.86

per km for the Kampala-Mombasa backhaul, 36 to 48% of the import rate (Heirstone analysis)

8 c

per tonne-km, Mombasa-Kampala road freight price in 2007; 6.4 to 8.6 c on 2025 rates (Heirstone analysis)

Waiting time uses land. A container that waits five days occupies a stacking slot, and an importer who cannot rely on the port holds more stock. Empty-container depots and export consolidation yards inland, at Kampala, Eldoret or Kigali, are corridor uses in their own right.



CHAPTER 3

03

The capacity overhang

Corridor finance is announced in capacity: tonnes a year of rail, TEU a year of terminal, hectares of dry port. Measured against what they carry, several of the past decade's projects use a small fraction of what they were built for.

The Addis Ababa-Djibouti railway, 753 km long and built for US$3.4bn to 4.2bn depending on the source, was designed for 10 to 20 million tonnes a year and carried about 3.2 million in 2025, according to Kana Television; the Ethiopian Railways Corporation had expected 6 to 7 million tonnes a year in its first years. TAZARA, built in 1975, carried 210,161 tonnes in the year to June 2022, about a third of its 600,000-tonne break-even volume, according to the operator. Kenya's Naivasha inland container depot, expanded to 450,000 TEU a year, handled 8,170 TEU in 2025; Kenyan press reported that regional traders preferred cheaper road haulage from Mombasa.

The assets that work are close to full. Nairobi's depot at Embakasi handled 311,647 TEU in 2025, about 69% of its stated capacity of over 450,000 TEU. Lomé moved 2.4m TEU in 2025 on a rated 2.7m, and Durban 2.54m in 2023 on about 2.9m. Nairobi's depot sits inside the market it serves; Naivasha's is 120 km further up the line and competes with trucks running straight from Mombasa.

For an investor buying land beside a new line or depot, the first questions are how much of the capacity is used, and why.

> *Designed for up to 20 million tonnes a year, the Addis Ababa-Djibouti railway carried about 3.2 million in 2025.*

16-32%

of design capacity used on the Addis Ababa-Djibouti railway in 2025, on a design range of 10 to 20 million tonnes (Heirstone analysis)

35%

of TAZARA's 600,000-tonne break-even volume carried in the year to June 2022 (Heirstone analysis)

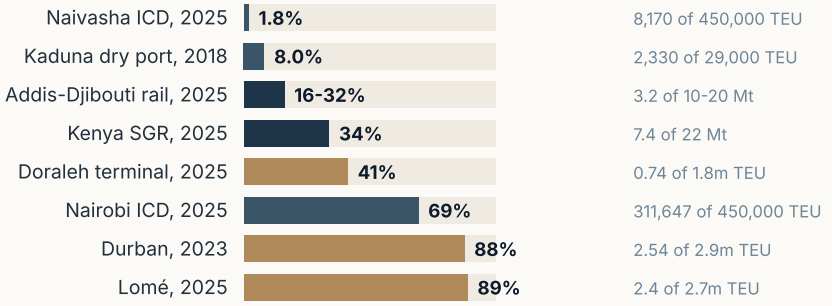
CAPACITY IN USE

Rail-served depots run from 2% to 69% full, and empty trains are expensive

EXHIBIT 7

The Heirstone Capacity Use Ledger: the same corridors hold full ports and near-empty depots

Latest throughput as a share of design or rated capacity, by asset, %. Gold: ports; blue: rail lines; slate: inland depots. Addis-Djibouti: bar on the 20 Mt top of its design range; 32% on the 10 Mt bottom

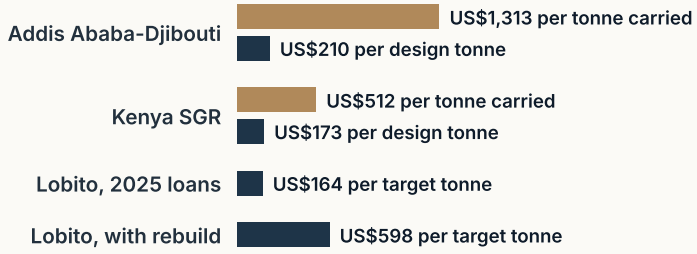


Source: Heirstone analysis of NCTTCA (2026), Kenya Ports Authority, Kenya Railways, SGTD (Sep 2025), Agence Ecofin (May 2026), BusinessDay Nigeria (Feb 2019), Lloyd's List via Maritimafrica (2024); Addis-Djibouti tonnage from Kana Television (Jul 2026) and design range from Uganda's SGR project (2017); Kenya SGR design from Railway Technology (2020); Durban capacity from The Maritime Executive (2021).

EXHIBIT 8

On 2025 tonnage, the Addis Ababa-Djibouti railway's construction cost comes to US$1,100 to 1,300 per tonne carried

Construction or financing cost divided by tonnes carried in 2025 (gold) and by design or target capacity (blue), US$ per annual tonne. Addis Ababa-Djibouti at US$4.2bn and 20 Mt; at the US$3.4bn cited by the US government, about US$1,100 per tonne carried. Lobito with rebuild adds the 2006-14 rebuild, put at about US$2bn



Source: Heirstone analysis of Djibouti Ports and Free Zones Authority (2017), US International Trade Administration (2022), Uganda SGR project (2017), Kana Television (2026), Railway Technology (2020), Lobito Corridor Transit Transport Facilitation Agency, NCTTCA (2026) and Railway Gazette International (Dec 2025). Lobito is measured on target tonnage, as actual 2024-25 tonnage is not published.

The gap between cost per tonne carried and cost per tonne of capacity is the cost of empty trains. On the Ethiopian line it is a factor of three to six; on Kenya's SGR, which carried 7.425 million tonnes in 2025, about three.

Rail frequency, more than land or sheds, sets the ceiling. Tanzania's Isaka dry port is served by three metre-gauge trains a day, which caps its throughput, according to Billionaires.Africa; Doraleh's rail terminal in Djibouti can handle five to six.

3-6x

cost per tonne carried against cost per design tonne, Addis Ababa-Djibouti railway, 2025, on its 10 to 20 Mt design range (Heirstone analysis)

3

metre-gauge trains a day serving Isaka dry port, Tanzania

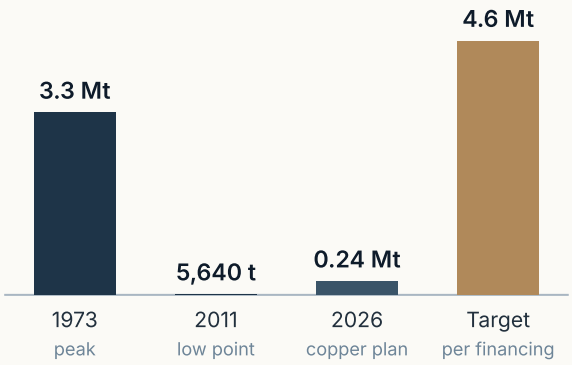
RESTORATION AND NEW BUILD

Lobito restores a 1970s corridor; state dry ports take years to fill

EXHIBIT 9

Lobito's target is 1.4 times what the Benguela Railway carried at its 1973 peak

Tonnes carried or targeted on the Lobito line; copper volume projected from Kolwezi via the line in 2026



Source: Lobito Corridor Transit Transport Facilitation Agency (1973; 2011, citing Jornal de Angola, May 2012); IRIS (Nov 2025, copper projection); Railway Gazette International (Dec 2025, target).

US$753m

DFC and DBSA financing for Lobito Atlantic Railway, financial close 17 December 2025

28%

completion of the 732 ha Ferkessédougou dry port in April 2026, five years after works began

2,734 ha

of state-led corridor land at Kribi, Ferkessédougou and Kwala (Heirstone analysis)

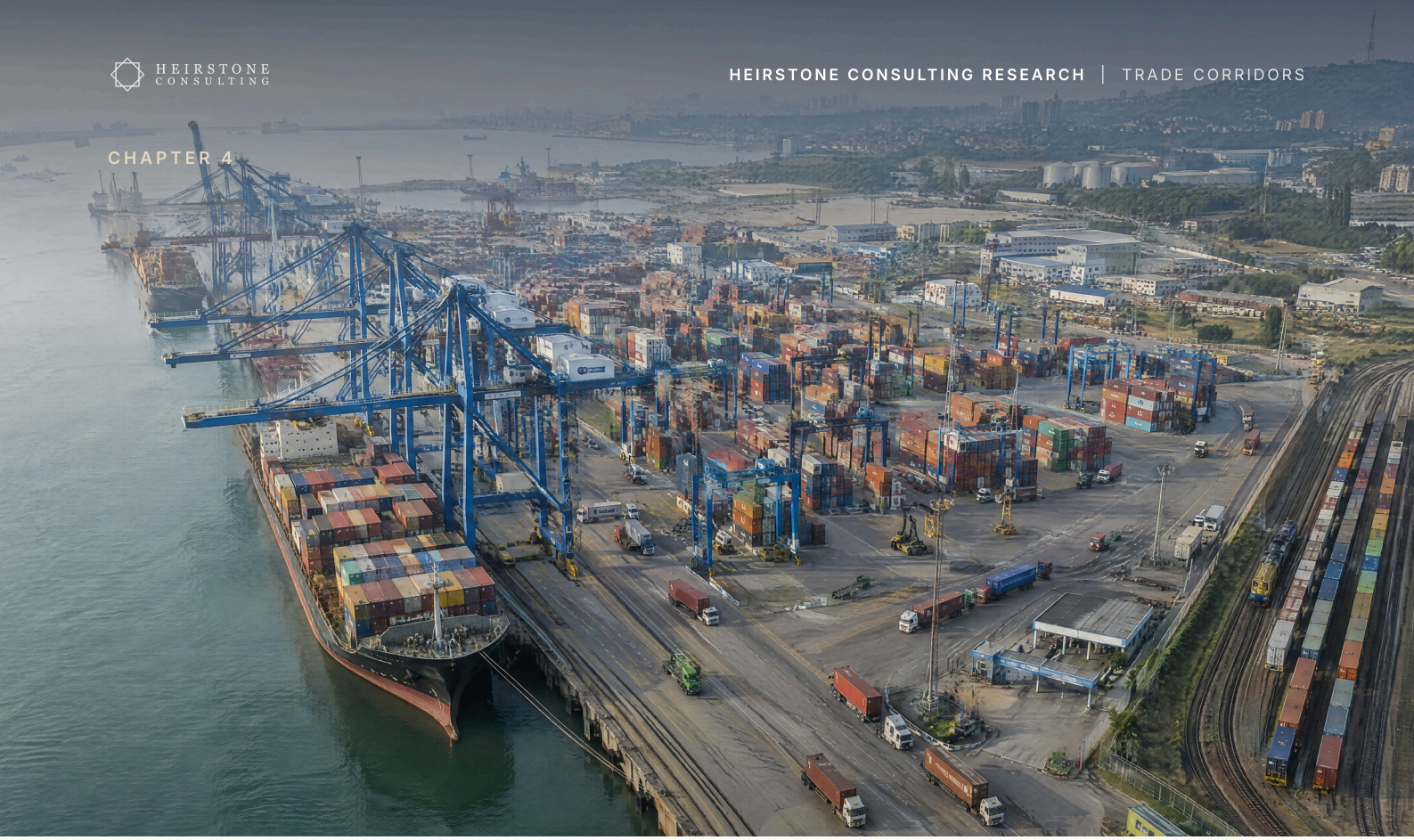
Lobito Atlantic Railway reached financial close on US$753m in December 2025: US$553m from the US International Development Finance Corporation and US$200m from the Development Bank of Southern Africa, on top of the concession commitments of US$455m in Angola and up to US$100m in the DRC announced in 2023. IRIS counts more than US$10bn of public and private commitments along the corridor. The financing targets 4.6 million tonnes a year. The Benguela Railway carried 3.3 million tonnes in 1973 and 5,640 tonnes in 2011, so the target is 1.4 times the line's historic peak: a restoration of the 1970s corridor more than the creation of a new one.

For land, that argues for patience at Benguela and Huambo and attention at the two places where cargo must stop. One is the port of Lobito, where Africa Global Logistics took the multipurpose and container terminal concession in December 2023 with nearly €100m (about US$108m at the 2023 average rate) of planned investment. The other is the Luau-Dilolo crossing into the DRC. Ivanhoe's Kamoa-Kakula has agreed to ship 120,000 to 240,000 tonnes of copper a year on the line; IRIS notes there is still no published tariff per tonne, so the promised saving cannot yet be tested.

State-built dry ports carry the same risk on a smaller scale. Côte d'Ivoire launched the 732 ha Ferkessédougou dry port in May 2021 at a cost of FCFA 254.2bn (about US$474m as reported at launch); in April 2026 the Transport Ministry put it at 28% complete. Nigeria's Kaduna dry port handled 2,330 TEU in its first year against a stated minimum capacity of 29,000, with exports held back by a lack of locomotives and wagons.

Three state-led corridor land banks with stated areas, Kribi's integrated industrial zone (1,500 ha), Ferkessédougou (732 ha) and Kwala (502 ha), add up to 2,734 ha. Supply on that scale, delivered by the state, is likely to take years to absorb, and occupiers are likely to go first to the nodes where trains and customs already work.

CHAPTER 4



04

Gateways, hubs and the towns behind the border

Africa's best-run container ports are transhipment hubs in the north. The gateways that serve inland markets are slower, busier and short of land.

The Container Port Performance Index of the World Bank and S&P Global measures how long ships spend in port. In the 2023 edition Tanger Med ranked 4th of 405 ports, the only African port in the global top ten, and Port Said 16th. In the 2024 edition, released in September 2025, Port Said was 3rd and Tanger Med 5th, as reported by Agence Ecofin. Most Sub-Saharan gateways that serve inland markets ranked between 309th (Lagos) and 405th (Cape Town) in 2023; Berbera, at 106th, was the exception.

Their volumes have grown regardless. Tanger Med handled 11.1m TEU in 2025, 2.3 times its 2019 throughput and about 15% a year compound (Heirstone analysis). Abidjan almost doubled to 1.6m TEU between 2022 and 2024 after its second container terminal opened. Mombasa grew 30% between 2023 and 2025, to 2.11m TEU, and Lomé reached 2.4m TEU in 2025.

Behind the gateways, the towns where trucks and trains stop are growing faster than the capitals they serve. This chapter looks at both ends: the port and the border.

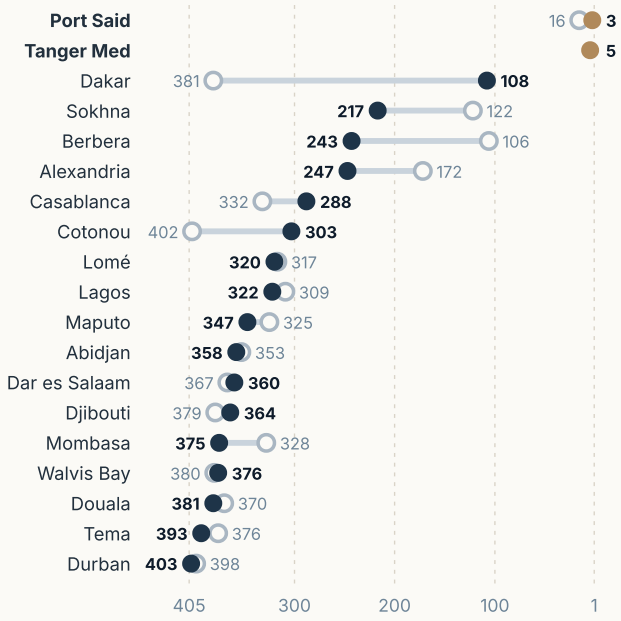
TWO PORT ECONOMIES

The hubs lead the world on efficiency; the gateways grow on volume

EXHIBIT 10

Two North African hubs rank in the global top five; most Sub-Saharan gateways rank below 300th

Container Port Performance Index world rank, 2023 (ring) and 2024 (dot) editions; better ranks to the right. In 2024 Sokhna and Alexandria rank 4th and 6th of the 19 ports shown



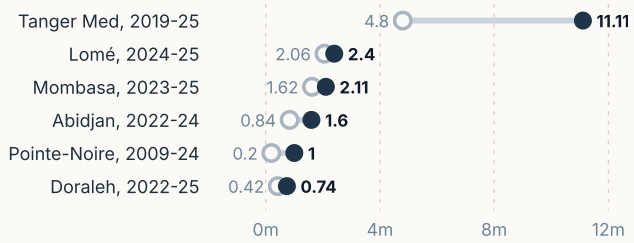
Source: World Bank and S&P Global, Container Port Performance Index 2023 (2024), Table E.1; CPPI 2024 ranks as published by Agence Ecofin (23 Sep 2025). Djibouti's authorities rejected the 2023 ranking.

Index ranks move sharply between editions. Dakar went from 381st to 108th, and Djibouti from 29th in 2022 to 379th in 2023, a ranking its port authorities rejected, and 364th in 2024. Investment cases should use several editions. Ownership is steadier. Africa Global Logistics, part of the MSC group, runs or holds stakes at Abidjan, Cotonou, Tema, Pointe-Noire, Lobito and the Walvis Bay bulk terminal; DP World at Dakar and the new Ndayane port, Berbera, Dar es Salaam and Maputo; Abu Dhabi's AD Ports Group and India's Adani hold Dar es Salaam's second container terminal.

EXHIBIT 11

Volumes have grown fast at hubs and gateways alike

Container throughput, million TEU, earliest (ring) and latest (dot) year available; Pointe-Noire latest is over 1m a year in 2022-24. Port Said (3.9m to 4.4m TEU in 2024, sources differ) is not shown, as no earlier figure could be verified



Source: Tanger Med Port Authority (2020, 2026); Le360 Afrique; KPA via Kenya News Agency and Maritimafrica; Agence Ecofin (Feb 2025); Maritimafrica (Jul 2025); SGTD (Sep 2025, units read as thousand TEU). Growth by Heirstone.

The two groups create different kinds of land value. A transhipment hub's boxes mostly stay on the quay, so its land play is a free zone that turns port efficiency into manufacturing and distribution: Tanger Med Zones runs a 3,000 ha platform with about 1,500 companies and 145,000 jobs. A gateway's cargo has to leave the port, so its land play is relief: container freight stations, empty depots, truck parks and rail links close to the gate.

Land behind the gate will be contested between these operators' logistics arms and local developers. Transnet's August 2026 call for proposals on 25-year leases to build and run logistics facilities on about 7 ha and 10 ha at Durban's Bayhead shows one way a port owner can open it to private capital. The Gulf appears in this story mainly as an operator and investor: DP World and AD Ports between them hold concessions at four of the gateways in this report, and DP World is building a fifth at Ndayane.

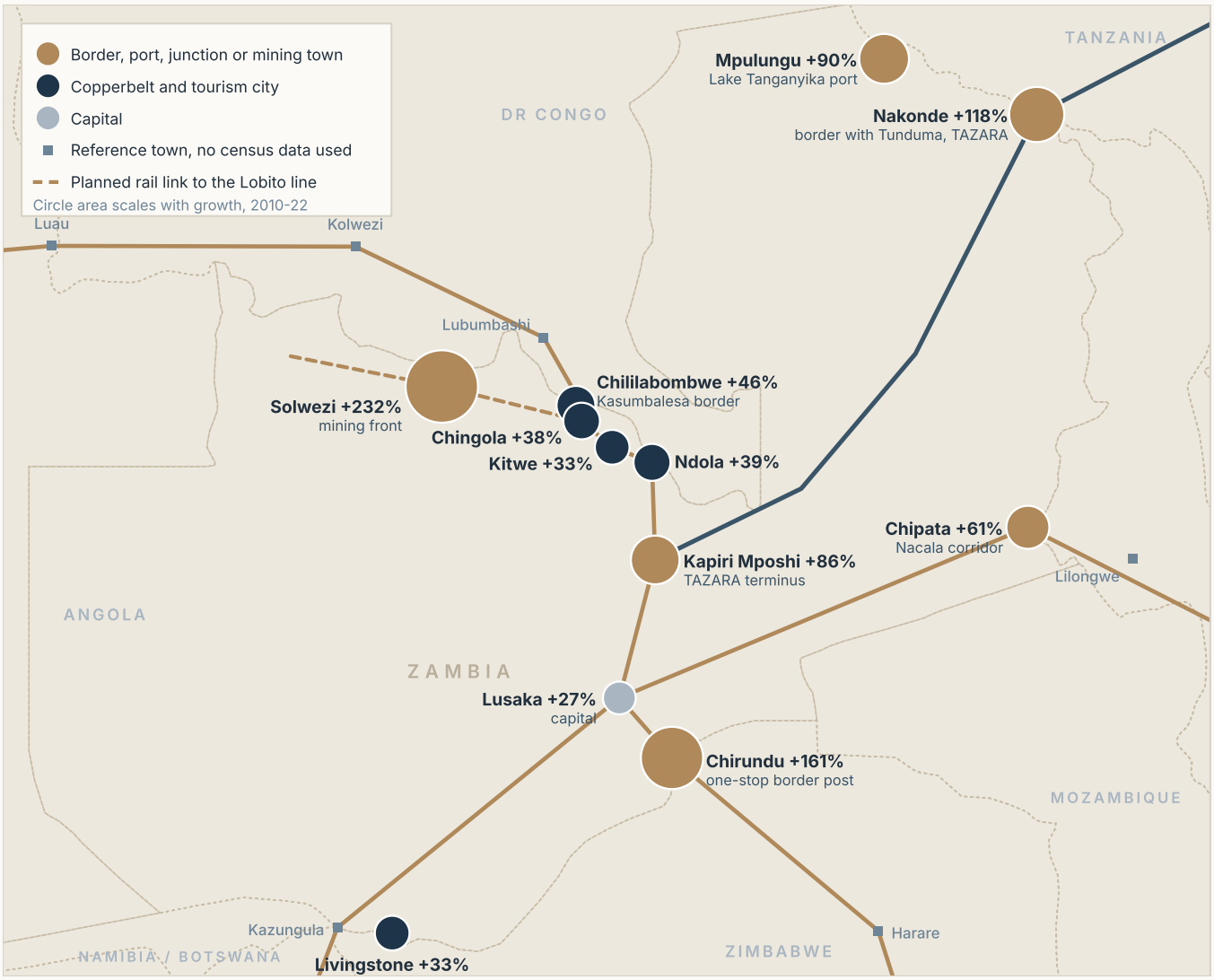
THE TOWNS BEHIND THE BORDER

Zambia's border and junction towns grew up to six times as fast as Lusaka

EXHIBIT 12

Between the 2010 and 2022 censuses, Chirundu grew 161% and Nakonde 118%; Lusaka grew 27%

Population change between the 2010 and 2022 censuses, Zambian towns on the North-South, TAZARA, Nacala and Copperbelt corridors, %. Corridors drawn schematically between approximate town centres



Source: Zambia Statistics Agency censuses via City Population (Thomas Brinkhoff); growth rates by Heirstone. Town positions are approximate GeoNames centroids; routes are schematic. Planned link per IRIS (Nov 2025).

Census data are the clearest public signal of corridor real estate. Chirundu, on the one-stop border post with Zimbabwe, grew from 13,585 people in 2010 to 35,403 in 2022. Nakonde, facing Tunduma on the Tanzanian border and the TAZARA line, grew to 91,107. Kapiri Mposhi, where TAZARA meets the Zambian network, grew 86%. Solwezi, the mining town on the planned rail link to the Lobito line, grew 232%, about six times Chingola's rate (Heirstone analysis).

The pattern holds beyond Zambia. Beitbridge in Zimbabwe, the busiest road border in Southern Africa, grew from 11,596 people in 1992 to 58,574 in 2022, about five-fold, while Harare grew 25%. The bases are small and population is an imperfect proxy, but no public land-price series exists for any African border town, and growth of this order means demand for plots, sheds and housing.

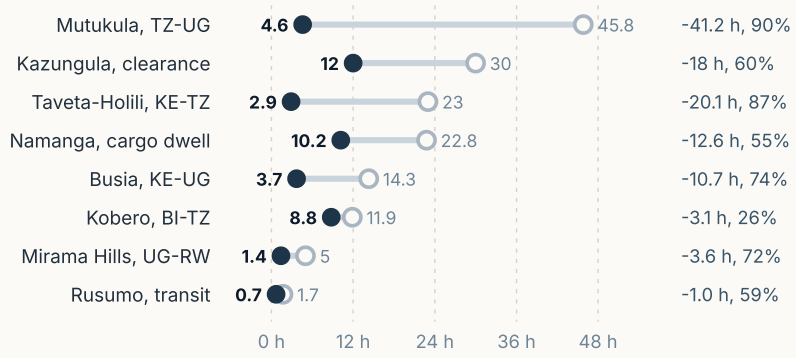
FASTER BORDERS

One-stop border posts save hours, and the towns still grow

EXHIBIT 13

Published surveys show one-stop posts cutting crossings by 26 to 90%

Truck crossing, clearance or dwell time before (ring) and after (dot) a one-stop border post, hours; direction or measure as stated; survey years differ (2011 to 2023)



Source: TradeMark Africa OSBP portal; Central Corridor TTFA (2015-16); JICA (2026, Namanga); Freight News (Aug 2023, Kazungula). Hours saved and percentages by Heirstone where not published.

Chirundu, Africa's first one-stop post, opened in December 2009 and cut crossings from 2 to 9 days to most vehicles cleared within a day. The town still grew 161% over the next twelve years, while traffic outgrew the post: 100 trucks a day in 2004, 260 in 2010 and 470 in 2011, then queues of up to 5 km from 2013 and waits above 36 hours reported in 2019, according to a conference presentation on the post. Kasumbalesa, on the DRC-Zambia Copperbelt border, had a 72 km queue in April 2019, with trucks moving about 2 km a day, according to Freight News.

A post that trucks pass through quickly supports fuel, food and clearing agents. A town where cargo is stored, moved between modes or gathered for the next leg supports bonded warehouses, truck parks, staging yards and housing.

Investors should look at the second kind. As posts get faster, demand moves from the parking bay to the bonded shed and the clearing office, and it moves towards towns with a rail transfer or a large consuming market. Nakonde-Tunduma, where the TAZARA line and the main road both cross into Zambia, has both.

470

trucks a day at Chirundu in 2011, up from 100 in 2004; waits were back above 36 hours by 2019

72 km

queue at Kasumbalesa on the DRC-Zambia border in April 2019, with trucks moving about 2 km a day

5x

growth of Beitbridge between the 1992 and 2022 censuses; Harare grew 25%

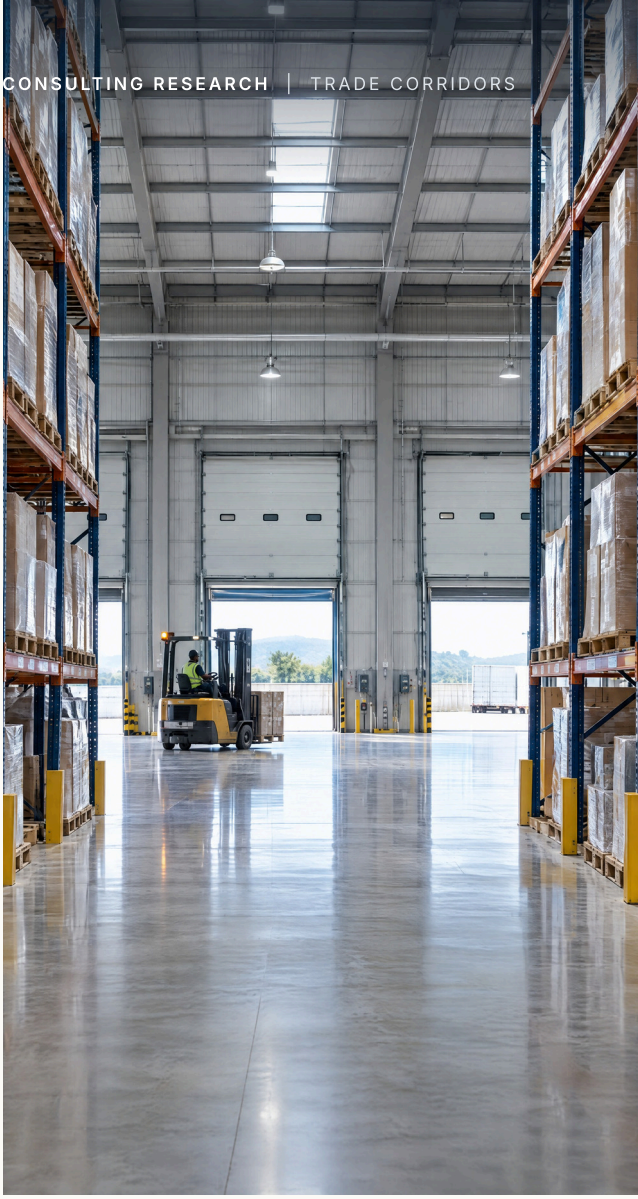
US$259m

cost of the Kazungula bridge (2021); transporters reported Johannesburg-Lusaka runs of about three days against four via Beitbridge

CHAPTER 5

05

Flat rents, scarce land



Since 2021, prime warehouse rents in US dollars have been flat or lower in most corridor cities. The value created at corridor nodes is showing up in land and in yields.

Knight Frank's H2 2025 dashboard puts prime logistics rents in Africa's main corridor cities in a narrow band: US$7 per m^2 a month in Kampala, 6.5 in Lagos, 6 in Nairobi, 5.5 in Johannesburg, 5 in Dar es Salaam, 4.5 in Lusaka and 4 in Cairo, the lowest of these seven headline rents for occupiers paying in dollars. Against the firm's H1 2021 figures, only Lagos (+48%) and Kampala (+17%) have risen in dollars; Nairobi, Johannesburg and Lusaka are unchanged, and Dar es Salaam (-29%) and Cairo (-20%) are lower.

Currency explains part of this. In Egyptian pounds, Cairo's prime rent rose about 2.5 times between 2021 and 2025 (Heirstone analysis at World Bank average rates), and the Tanzanian shilling moved from about 2,298 to 2,598 to the dollar between 2021 and 2024. Most international investors and lenders underwrite in hard currency, and in hard currency the corridor upgrades of the past five years have not yet produced rental growth.

Occupancy has tightened all the same. Knight Frank's continental average for modern warehouses rose from 75% to 83% in the year to H1 2025, and vacancy in South African logistics buildings over 20,000 m^2 was 0.5% in early 2024, according to JLL. The next two pages show where that pressure is turning into value: in yields that price distance from the sea, and in serviced land at the nodes that work.

+48%

Lagos prime logistics rent in US$, H1 2021 to H2 2025, the largest rise among the cities in Exhibit 14

-29%

Dar es Salaam prime rent in US$ over the same period, the largest fall among those cities

83%

average occupancy of modern warehouses in Knight Frank's African cities, H1 2025

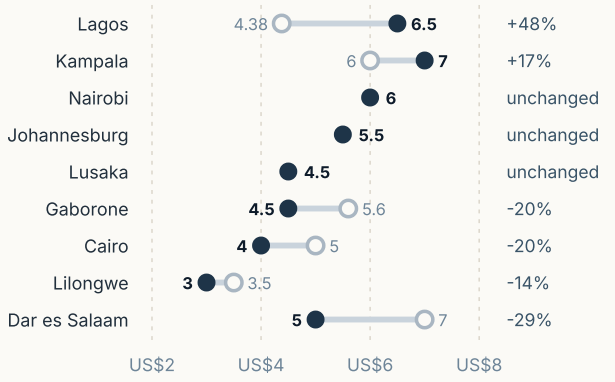
RENTS AND YIELDS

Distance from the sea shows up in the yield

EXHIBIT 14

In US dollars, only Lagos and Kampala prime rents are above their 2021 level

Prime logistics rent, US$ per m² a month, H1 2021 (ring) and H2 2025 (dot); Lusaka H2 2025 is the top of a US$3 to 4.5 range; in Egyptian pounds, Cairo's rent rose about 2.5 times



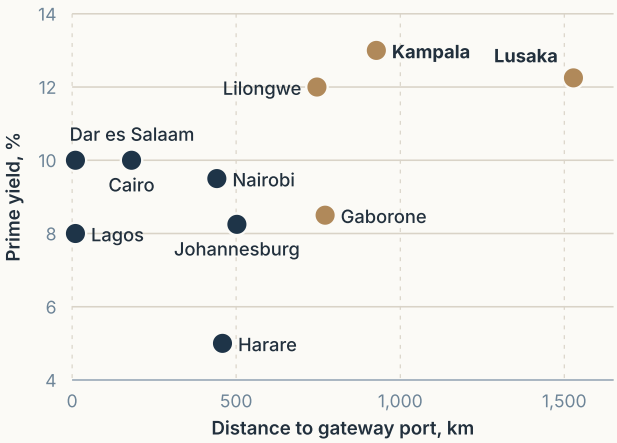
Source: Knight Frank, Africa Logistics & Industrial Review H2 2021; Africa Industrial Market Dashboard H2 2025; changes by Heirstone.

Occupiers who pay in local currency, and competition from older stock, are likely to cap rents. Lusaka's secondary space lets at US$1.50 to 2.50, and within cities the spread is wider than between them: Cairo runs from US$3 in Sadat City and 10th of Ramadan to US$8 in 6th of October, and Lagos from US$3 in Ikeja to US$6.5 at Alaro City. On these figures the location within a city moves rent more than the choice of city.

EXHIBIT 15

The two cities furthest from the sea carry the highest prime yields

Prime logistics yield, H2 2025, %, against straight-line distance to the assumed gateway port, km; gold: cities more than 700 km inland. Ten cities, so the relationship is indicative. Harare has about 40% industrial vacancy



Source: Knight Frank, Africa Industrial Market Dashboard H2 2025; distances and correlation by Heirstone (gateways assumed: Mombasa, Dar es Salaam, Durban, Beira, Nacala, Alexandria, Lagos).

Across the ten Knight Frank cities, distance from the gateway port correlates at about +0.5 with prime yield and about -0.1 with rent (Heirstone analysis; +0.6 without Harare). Ten points are too few for a firm conclusion, so we read this as indicative. Kampala, 927 km inland, prices prime warehouses at 13% and Lusaka, 1,528 km, at 12 to 12.5%, against 8% in Lagos and 8.25% in Johannesburg. The premium is consistent with investors pricing currency, liquidity and supply risk further inland.

> *On a stable rent, a fall in prime yield from 13% to 12% adds about 8% to capital value (Heirstone analysis). For an inland landlord, that is what a faster corridor could be worth if it narrows the inland premium.*

LAND

Serviced land at a working node is the scarce asset

EXHIBIT 16

Where serviced land sits at a working node it is full; where land was only designated, it waits

Take-up, occupancy and delivery indicators for corridor land and logistics stock, latest available

NODE OR ZONE	COUNTRY	MEASURE	LATEST	DATE	READING
Lusaka South MFEZ	Zambia	Industrial land allocated	548.6 of 548.6 ha; 20 investors waiting	Jun 2025	**Full**
Dongo Kundu SEZ, Mombasa	Kenya	Land applied for against land available	about 7,000 vs 3,000 acres (2.3x)	2026	**Oversubscribed**
6th of October City, Cairo	Egypt	Industrial occupancy	95%	H1 2025	**Full**
ALP North, Tatu City	Kenya	Occupancy after a 20,000 m^2 letting	90%	Q1 2023	**Full**
Logistics buildings over 20,000 m^2	South Africa	Vacancy	0.5%	Q1 2024	**Full**
Modern warehouses, Knight Frank cities	Africa	Average occupancy	83% (75% a year earlier)	H1 2025	**Tightening**
Akoupé-Zeudji PK24, Abidjan	Côte d'Ivoire	Land allocated; firms operating	284 of 1,000 ha; 67 of 173 firms	Feb 2026	**Filling**
Gazetted SEZs (36+)	Kenya	Jobs created against projection	4,392 vs 26,000 (17%)	early 2024	**Lagging**
Ferkessédougou dry port	Côte d'Ivoire	Construction progress	28% complete	Apr 2026	**Lagging**
Naivasha ICD	Kenya	Capacity in use	1.8%	2025	**Idle**

Source: IDC Zambia (Aug 2025); Capital FM (29 Jul 2026, Kenyan SEZ figures, single source); Knight Frank dashboards (Q1 2023, H1 2025); JLL via BusinessTech (Aug 2024); Fraternité Matin (Feb 2026); Financial Afrik (Apr 2026); NCTTCA (2026), Kenya Railways; Heirstone calculations.

Lusaka South Multi-Facility Economic Zone allocated the last of its 548.6 ha of industrial land by June 2025, with 140 approved investors, 31 operating, US$1.58bn invested and 9,804 jobs. Mombasa's Dongo Kundu zone received 97 applications for nearly 7,000 acres against about 3,000 available, according to Capital FM. In Durban, Cushman & Wakefield | Broll reports no meaningful release of flat, flood-free land in more than a decade, and A-grade distribution centres let at R105 to 110 per m^2 triple net, about US$5.9 to 6.1 at the 2025 average rate.

Zones that were designated without serviced land have filled more slowly. According to Capital FM, Kenya has gazetted at least 36 special economic zones, which by early 2024 had attracted US$960.7m and 4,392 jobs against 26,000 projected; the same report credits Tatu City, a privately serviced zone north of Nairobi, with more than 25,000 jobs, on a basis it does not reconcile with the national total. The private, port-integrated Lagos Free Zone cites US$2.7bn of employed capital.

The stock gap is large but depends on definitions. Morocco's logistics association counts more than 20 million m^2 of warehousing in about 13,000 buildings, about 0.53 m^2 per person. Knight Frank put Nigeria's modern stock at about 300,000 m^2 in 2021, about 0.0014 m^2 per person, against a requirement of about 1 million m^2 (Heirstone analysis, World Bank population). Estate Intel finds up to 270,000 m^2 at a single Lagos node, Oba Akran, so the modern estimate is narrow. Even at ten times that estimate, Nigeria would hold about a fortieth of Morocco's stock per head.

100%

of Lusaka South MFEZ's 548.6 ha allocated by June 2025, with 20 investors waiting

2.3x

land sought against land available at Dongo Kundu SEZ, Mombasa (Heirstone analysis)

17%

of projected jobs created by Kenya's gazetted SEZs by early 2024, on Capital FM figures (Heirstone analysis)

CHAPTER 6

06

The Heirstone Corridor Node Benchmark

We assembled public data on 34 corridor nodes in 22 countries: transhipment hubs, gateway ports, rail-served inland depots, border towns and junction towns. For each we record throughput, capacity in use, the hours it saves or loses, and the best available real estate signal.

The five node types capture corridor value in different ways and support different real estate. Exhibit 17 sets out the typology, Exhibit 18 maps the nodes, Exhibit 19 gives the data and Exhibit 20 scores sixteen of them on the Heirstone Node Test.

EXHIBIT 17

The Heirstone Corridor Node Typology: five ways a corridor creates land value

Node types, the mechanism by which each captures value, the real estate it supports and the evidence in this report

NODE TYPE	EXAMPLES	HOW VALUE IS CREATED	REAL ESTATE IT SUPPORTS	EVIDENCE	WATCH FOR
Transhipment hub	Tanger Med, Port Said; Lomé and Djibouti in part	Port efficiency attracts exporters to adjoining free zones; most boxes never leave the quay	Free-zone industrial plots, export processing and automotive supply parks, ro-ro and truck terminals	Tanger Med: CPPI 4th (2023), 11.1m TEU (2025), zones of 3,000 ha with about 145,000 jobs	Thin domestic hinterland; land controlled by the port authority
Gateway port	Mombasa, Dar es Salaam, Durban, Abidjan, Lagos, Tema, Dakar, Maputo	Cargo must leave the port; every hour of dwell consumes yard, parking and storage land	Container freight stations, empty depots, truck parks, port-linked SEZs, city distribution centres	Mombasa dwell 106.7 h; Dongo Kundu sought 2.3 times over; Durban has had no flat land release in a decade	Operators' own logistics arms take the best land; congestion moves inland
Rail-served inland depot	Nairobi ICD, Modjo, Kwala; Naivasha, Kaduna and Ferkessédougou as cautions	Customs clearance moves inland to the railhead, so stock is held where it is cleared	Bonded warehousing, regional distribution centres and light industry beside the depot	Nairobi ICD at 69% of capacity against Naivasha at 1.8%; Modjo handles over 78% of Ethiopia's imports	Too few trains a day; road stays cheaper; customs not on site
Border town	Chirundu, Kasumbalesa, Nakonde-Tunduma, Beitbridge; Malaba and Namanga as pass-through cases	Trucks wait, clear and transfer; the town grows if cargo breaks bulk or a market sits behind it	Truck parks, bonded yards, clearing-agent offices, workforce housing, retail	Chirundu +161% and Nakonde +118% against Lusaka +27% (2010-22); Beitbridge five-fold since 1992	A faster post cuts waiting demand; growth can stall at pass-through posts
Junction and mining-front town	Kapiri Mposhi, Solwezi, Lusaka South, Dire Dawa, Naivasha	Modes meet or production starts; staging and processing need space	Serviced industrial land, staging and marshalling yards, housing for workers and contractors	Solwezi +232% (2010-22); Lusaka South fully allocated with 20 investors waiting	Single-commodity cycles; rail links that are planned but not financed

Source: Heirstone Consulting typology; evidence as cited in Chapters 2 to 5.

A node can change type. Djibouti is turning from Ethiopia's gateway into a hub, with transhipment capacity planned to rise from 690,000 TEU to 2.9m TEU by 2030, and Kwala is meant to move clearance for the Central Corridor from Dar es Salaam's port to a railhead. The type tells an investor which product to build and which risk to price.

THE BENCHMARK NODES

The benchmark nodes and the corridors that link them

EXHIBIT 18

The value sits at the nodes: ports, depots, border towns and junctions on the main corridors

Benchmark nodes and other ports and towns on the main corridors, by type; Busia and Solwezi are omitted for legibility



Source: Heirstone Corridor Node Benchmark 2026; corridor definitions from corridor agencies, AfDB, IRIS and the research cited in this report. Positions are approximate town or port centres (GeoNames centroids and Heirstone geocodes); routes are drawn as straight lines between towns and do not show alignments. The Egypt-Sudan and Berbera corridors are shown dashed because current volumes and status were not verified.

THE BENCHMARK

Thirty-four nodes, one table

Throughput and capacity as published; CPPI ranks are World Bank 2023 / 2024 editions (of 405 in 2023). Town growth is between census years. n/p: not published in the sources used. n/a: not applicable. DB2020: World Bank Doing Business 2020.

EXHIBIT 19

The Heirstone Corridor Node Benchmark 2026

NODE	COUNTRY	CORRIDOR	THROUGHPUT, LATEST	CAPACITY IN USE	TIME SIGNAL	REAL ESTATE SIGNAL
TRANSHIPMENT HUBS						
Tanger Med	Morocco	Tanger Med to Europe and West Africa	11.1m TEU, 161 Mt (2025); +8.4% TEU	n/p	CPPI 4th (2023), 5th (2024); import clearance 57 h (DB2020)	Zones: 3,000 ha, about 1,500 firms, 145,000 jobs
Port Said (East)	Egypt	Suez Canal	3.9 to 4.4m TEU (2024; sources differ)	SCCT capacity 7.0m TEU after Nov 2025	CPPI 16th (2023), 3rd (2024)	US$500m SCCT expansion (2025)
Lomé	Togo	Lomé-Ouagadougou; Abidjan-Lagos	2.4m TEU (2025); +16%	89% of 2.7m TEU	CPPI 317th / 320th; Sahel shippers' costs cut 15-25% (Ecofin)	92% of transit bound for Mali, Burkina Faso, Niger
Djibouti (Doraleh)	Djibouti	Djibouti-Addis Ababa	741k TEU (2025); +76% on 2022	41% of 1.8m TEU	5-6 trains a day; CPPI 379th / 364th	Transhipment capacity plan 2.9m TEU by 2030
GATEWAY PORTS						
Mombasa	Kenya	Northern	2.11m TEU, 45.45 Mt (2025)	n/p	Dwell 106.7 h against 48 h target	Dongo Kundu SEZ: about 7,000 acres sought, 3,000 available
Dar es Salaam	Tanzania	Central; TAZARA	27.7 Mt (FY2024/25); +17%	CT2: 0.82m of 1.0m TEU (2023)	Ro-ro discharge 300+ h to under 28 h (DP World)	8 of 12 berths privately operated
Durban	South Africa	North-South	2.54m TEU (2023)	About 88% of about 2.9m TEU	CPPI 398th / 403rd	No flat land release in a decade; 25-year Bayhead leases offered
Abidjan	Côte d'Ivoire	Abidjan-Ouagadougou; Abidjan-Lagos	1.6m TEU (2024); 46.6 Mt (2025)	TC2 rated 1.5m TEU	CPPI 353rd / 358th; Mali transit +76% (2025)	PK24: 284 of 1,000 ha allocated
Tema (MPS terminal)	Ghana	Tema-Ouagadougou; Abidjan-Lagos	About 2.3m TEU (2025 projection)	Target capacity 3.0m TEU	CPPI 376th / 393rd	About 30% of Burkina Faso's imports
Lagos (Apapa, Tin Can, Lekki)	Nigeria	Abidjan-Lagos	Nigeria 2.1m TEU, 129.3 Mt (2025)	Lekki phase 1: 1.2m TEU	CPPI 309th / 322nd (Lagos)	Prime rent +48% since 2021; Lagos Free Zone US$2.7bn
Dakar	Senegal	Dakar-Bamako	881,289 TEU (2024); +10.1%	n/p	CPPI 381st to 108th	Ndayane: US$1.2bn port and zone, opening 2028
Maputo	Mozambique	Maputo	32.0 Mt (2025), a record	n/p	CPPI 325th / 347th	Concession since 2003 (MPDC)
Walvis Bay	Namibia	Walvis Bay corridors	n/p	750,000 containers a year	CPPI 380th / 376th	n/p
Lobito	Angola	Lobito	Rail target 4.6 Mt a year	1973 peak 3.3 Mt	Kolwezi in 8 days by rail; 25+ days by road to Dar	Terminal concession, nearly €100m
Kribi	Cameroon	Kribi	555,398 TEU (2025); 10.8 Mt (2024)	n/p	CPPI 366th / 397th	Industrial zone of 1,500 ha planned

Source: Heirstone Corridor Node Benchmark 2026, compiled from the sources listed for Exhibits 4 to 16 and in the notes.

THE BENCHMARK, CONTINUED

NODE	COUNTRY	CORRIDOR	THROUGHPUT, LATEST	CAPACITY IN USE	TIME SIGNAL	REAL ESTATE SIGNAL
RAIL-SERVED INLAND DEPOTS AND DRY PORTS						
Nairobi ICD	Kenya	Northern	311,647 TEU (2025)	69% of 450,000+ TEU	SGR and metre gauge	Tatu City 25,000+ jobs; ALP North 90% let
Naivasha ICD	Kenya	Northern	8,170 TEU (2025)	1.8% of 450,000 TEU	Traders prefer road	SEZ grew from 1,000 to 6,000 acres
Modjo	Ethiopia	Djibouti-Addis Ababa	Over 78% of Ethiopia's imports	136,038 TEU a year; sources differ	Rail to cut 3 days to 12 h (2017)	Eastern Industrial Zone: 110 firms, 16,750 jobs
Kwala	Tanzania	Central	Opened July 2025	300,000 boxes a year	SGR 45-60 min; road 30-35 h	502 ha plus industrial park
Isaka	Tanzania	Central	n/p	Capped by 3 trains a day	610 km by road to Kigali	Leased to a private operator
Kaduna	Nigeria	Lagos rail	2,330 TEU (2018)	8% of 29,000 TEU	Short of locomotives and wagons	n/p
Ferkessédougou	Côte d'Ivoire	Abidjan-Ouagadougou	Under construction	28% built (Apr 2026)	n/a	732 ha; FCFA 254.2bn
BORDER TOWNS AND POSTS						
Chirundu	Zambia, Zimbabwe	North-South	470 trucks a day (2011)	Waits above 36 h (2019)	2-9 days cut to under 1 day	Town +161% (2010-22)
Kasumbalesa	DRC, Zambia	Copperbelt	n/p	72 km queue (2019)	n/p	Chililabombwe +46%
Nakonde-Tunduma	Zambia, Tanzania	TAZARA; Dar es Salaam road	n/p	TAZARA at about 35% of break-even	72 h baseline; post from 2019	Nakonde +118% (2010-22)
Beitbridge	Zimbabwe, South Africa	North-South	Busiest road border in the region	n/p	About 4 days to Lusaka (2021)	Town five-fold since 1992
Kazungula	Zambia, Botswana	North-South	n/p	n/p	Clearance 30 h to 12 h; Lusaka in about 3 days	US$259.3m bridge (2021)
Busia	Kenya, Uganda	Northern	894 vehicles a day (2011)	n/p	14.3 h to 3.7 h (Kenya to Uganda)	n/p
Malaba	Kenya, Uganda	Northern	n/p	n/p	22.6 h baseline (2016)	n/p
Rusumo	Rwanda, Tanzania	Central	Corridor: about 80% of Rwanda's trade	n/p	Clearance about 4 h to 30 min	n/p
JUNCTION AND MINING-FRONT TOWNS						
Kapiri Mposhi	Zambia	TAZARA; North-South	TAZARA 0.21 Mt (FY2021/22)	About 35% of break-even	n/p	Town +86% (2010-22)
Solwezi	Zambia	Lobito (planned link)	n/p	Link works from Q3 2026 (plan)	n/p	Town +232% (2010-22)
Lusaka South MFEZ	Zambia	North-South; Dar; Lobito	US$1.58bn invested	548.6 ha, all allocated	n/p	20 investors waiting (2025)
Dire Dawa	Ethiopia	Djibouti-Addis Ababa	Dry port; 150 ha park	n/p	325 km to Djibouti port	Park 3 km from dry port

Source: Heirstone Corridor Node Benchmark 2026; population growth from national censuses via City Population; sources as for Exhibits 4 to 16.

The Heirstone Node Test: five questions, scored 0 to 2 from published evidence

1 STOP

Must cargo stop here for customs, a change of mode or a break of bulk? 2 for most cargo; 1 for some; 0 if it passes through.

2 USE

Is volume growing and capacity used? 2 if growing and over 60% used, or land fully let; 1 if growing with lower or unpublished use; 0 if under 20% of design or not operating.

3 HOURS

Does it save measured time? 2 for a top-50 CPPI port or a published saving of a day or more; 1 for a saving of hours, or one partly eroded; 0 if none is published.

4 REACH

Can it serve AfCFTA markets? 2 if the host and markets taking 80% of its hinterland flows have finalised schedules (US ITA, Dec 2025); 1 if some do; 0 if the host has not.

5 LAND

Is serviced land being taken up? 2 for full allocation, a waiting list or occupancy above 85%; 1 for partial take-up or indirect evidence; 0 if nothing is published.

THE NODE TEST

Five nodes reach eight points; the Sahel gateways fall on reach, Lobito and Naivasha on use

The benchmark shows what each node does. The Node Test asks whether its land is likely to gain value as AfCFTA trade grows, scoring the five questions on the previous page for a maximum of 10. A low score can mean that evidence is missing rather than that demand is weak, and every score can be re-run as new data are published.

EXHIBIT 20

The Heirstone Node Test: five nodes score 8 of 10; Lobito, Abidjan and Naivasha score lowest on present evidence

Scores out of 2 on each test (filled circles) and total out of 10; Strong 8 to 10, Mixed 6 to 7, Thin 5 or less

NODE	TYPE	STOP	USE	HOURS	REACH	LAND	SCORE	EVIDENCE	MAIN GAP
Tanger Med	Hub	●●	●○	●●	●●	●○	8	Strong	Capacity use and zone occupancy not published
Nairobi ICD and warehouse belt	Depot	●●	●●	○○	●●	●●	8	Strong	No published time saving
Port Said and Sokhna	Hub	●●	●○	●●	●●	●○	8	Strong	Capacity use and land take-up not fully published
Dar es Salaam and Kwala	Gateway	●●	●●	●●	●○	●○	8	Strong	DRC schedule not finalised
Modjo	Depot	●●	●○	●●	●●	●○	8	Strong	Capacity sources conflict
Mombasa and Dongo Kundu	Gateway	●●	●○	○○	●●	●●	7	Mixed	Dwell 106.7 h and rising
Durban	Gateway	●●	●●	○○	●○	●●	7	Mixed	CPPI 403rd (2024)
Chirundu	Border	●●	●●	●○	●○	●○	7	Mixed	Gains eroded by traffic
Kazungula	Border	●●	●○	●●	●●	○○	7	Mixed	No land or traffic data
Kasumbalesa	Border	●●	●●	○○	●○	●○	6	Mixed	72 km queue (2019)
Nakonde-Tunduma	Border	●●	●○	○○	●●	●○	6	Mixed	No post-OSBP survey
Lusaka South MFEZ	Junction	●○	●●	○○	●○	●●	6	Mixed	Land full; needs new phase
Lomé	Hub	●●	●●	●○	○○	○○	5	Thin	Togo schedule not finalised
Abidjan and PK24	Gateway	●●	●○	○○	○○	●○	4	Thin	Côte d'Ivoire not on list
Lobito	Gateway	●●	○○	●●	○○	○○	4	Thin	Only target tonnage published
Naivasha ICD and SEZ	Depot	●○	○○	○○	●●	●○	4	Thin	1.8% of capacity used

Source: Heirstone analysis of the Heirstone Corridor Node Benchmark 2026 (Exhibit 19), using the scoring rules on the previous page; AfCFTA schedule status from the US International Trade Administration (Dec 2025).

No node scores 9 or 10 on present evidence. Five score 8: Tanger Med, Port Said, Modjo, Dar es Salaam with Kwala, and Nairobi's depot. Each is a place where cargo must stop in a country with a finalised schedule. The three hubs and Modjo lose points on published use or land; Nairobi's depot has full land and a well-used depot but no published time saving.

The middle group fails one test badly. Mombasa and Durban lose on hours, because their ports are slow; Kazungula on land, because no data exist; Chirundu and Kasumbalesa on the erosion of their border gains. These are the nodes where a specific fix, a faster gate or a bonded yard, would add most value.

The bottom group needs reading with care. Lomé and Abidjan lose up to two points each because Togo, Côte d'Ivoire, Mali and Burkina Faso were not on the US list of finalised schedules in December 2025; a gazette notice in Lomé or Abidjan would move them up. Lobito scores 4 because only its target tonnage is published, and Naivasha because its depot is almost empty. For an investor, the test separates nodes to buy on evidence from nodes that are options on future evidence.



CHAPTER 7

07 North Africa and Sub-Saharan Africa

NORTH AFRICA

Morocco: the clearest case of port-led real estate

Tanger Med handled 11.1m TEU, 161 million tonnes, 535,203 TIR trucks and 526,862 vehicles in 2025; vehicle volumes, mostly Renault (327,569) and Stellantis (126,874), fell 12% on the year. Its zones cover 3,000 ha with about 1,500 companies and report US$20.61bn of business volume. Morocco's logistics association counts more than 20 million m^2 of warehousing, but 60% of sites are under 1,000 m^2 and cold storage covers about 4% of perishable volumes. The next product is large, modern and temperature-controlled space near the port and on the Kenitra-Casablanca axis. The association puts the threshold for viable Grade A space at Tanger Med at a rent below 45 dirhams per m^2 a month (about US$4.8 at the 2025 average rate) with occupancy above 85%. Morocco's schedule is finalised, and import clearance took 57 hours in Doing Business 2020, the fastest in our sample.

Egypt: a top-three port on efficiency, and canal-zone land

Port Said ranked 3rd in the 2024 port index, and APM Terminals' Suez Canal Container Terminal at East Port Said completed a US$500m expansion in November 2025 that took capacity to 7m TEU. Port Said and Alexandria are two of the four African ports in Lloyd's List's 2025 global top 100, at 53rd and 90th. The land opportunity lies in the Suez Canal Economic Zone ports, where a US$120m pharmaceutical hub agreement covers 96,828 m^2 at Sokhna, and in Greater Cairo, where prime rents ran from US$3 in Sadat City and 10th of Ramadan to US$8 in 6th of October City (95% occupied) in H1 2025. Egypt's finalised schedule and its LPI score of 3.1, joint second in Africa in 2023 and up from 2.37 in 2007, give it reach south and west. The road corridor to Sudan via Aswan and Wadi Halfa extends that reach, though its volumes are unpublished.

Algeria, Tunisia and Libya: thin data

No traffic counts are published for the Trans-Sahara Highway from Algiers to Lagos. Algiers had one of Africa's highest prime industrial rents in Knight Frank's 2021 survey (US$10 per m^2) and Tunis US$5 at a 14% yield; neither is covered by the firm's current dashboard. Algeria and Tunisia have finalised schedules. In Libya, a partnership announced in January 2026 aims to develop the Misurata Free Zone port as a 4m TEU transhipment hub, according to Maritimafrica.

11.1m

TEU through Tanger Med in 2025, 2.3 times its 2019 throughput

145,000

jobs on Tanger Med's 3,000 ha of zones

7m TEU

capacity at the Suez Canal Container Terminal, East Port Said, after November 2025

US$33.1bn

investment attracted to the Sokhna industrial zone of the Suez Canal Economic Zone across 547 projects by mid-2026, with more than 133,000 direct jobs (Egypt's Cabinet)

SUB-SAHARAN AFRICA: EAST AND SOUTHERN

In the east, the land answer sits at the port and at Nairobi; in the south, it is gateway-neutral

Kenya. Mombasa's dwell time ran 58.7 hours above target in 2025, and land is answering around the port and at Nairobi: Dongo Kundu is sought 2.3 times over and Nairobi's depot runs at 69%, while Naivasha's sits at 1.8%. Prime warehouse rent in Nairobi is US$6 at a 9.5% yield. ALP Industrial REIT listed in March 2026 after raising US$29.55m, according to Serrari Group, small beside South Africa's Equites at R28.7bn (about US$1.6bn at the 2025 average rate).

Uganda. Kampala takes 68.7% of Mombasa's transit cargo and has the highest prime rent (US$7) and yield (13%) among the cities Knight Frank tracks. Industrial land was valued at about US$1.5m an acre in 2021, above Nairobi's US$900,000. Namanve's park had 60% of contracted infrastructure complete in April 2025. A rail service reaching Kampala that worked like Nairobi's depot would compress that yield.

Tanzania. Dar es Salaam handled 27.7 million tonnes in FY2024/25, up 17%, with 8 of its 12 berths now privately run by DP World and a consortium of AD Ports Group and Adani. Kwala and its industrial park sit 45 to 60 minutes away by rail. The US$1.4bn TAZARA revitalisation agreed by China, Tanzania and Zambia in 2025, with a 30-year concession for a Chinese operator, would raise the importance of Nakonde-Tunduma and Kapiri Mposhi.

Rwanda. Import border compliance took 74 hours in Doing Business 2020, less than coastal Kenya (194) or Tanzania (402), and Rusumo truck clearance is down to about 30 minutes. The Central Corridor carries almost 80% of Rwanda's trade, which supports Kigali as a consolidation point for the Great Lakes.

Ethiopia and the Horn. Modjo dry port handles more than 78% of Ethiopia's imports, according to its operator ESL, and the Modjo-Adama-Dukem strip holds the private Eastern Industrial Zone (233 ha, 110 companies, 16,750 jobs) and the 365 ha Adama park. Djibouti's Doraleh terminal ran at about 41% of capacity in 2025 and plans 2.9m TEU of transhipment capacity by 2030; Berbera, ranked 106th in 2023 against Djibouti's 379th, is being developed by DP World with an economic zone along a corridor towards Ethiopia. Logistics land near Modjo serves either route.

South Africa. The continent's deepest logistics market holds 19.2% of intra-African trade, ranks 19th in the 2023 LPI and had 0.5% vacancy in large logistics buildings in early 2024; Cape Town A-grade rents rose about 58% in rand from Q2 2021 to R90 to 95 per m^2 (about US$5.0 to 5.3). Its ports are the weak link: Durban, with about 60% of the country's container traffic, ranked 403rd in 2024. A plan of about US$7bn announced in 2021 aims to take Durban from about 2.9m TEU to more than 11m over a decade, and ICTSI took 49% of Pier 2 from January 2026.

Zambia and the DRC. This is where the corridor contest is. Copper now leaves through Dar es Salaam, Walvis Bay, Beira and Durban, with smaller volumes via Nacala and Lobito, according to Freight News. That favours gateway-neutral assets: bonded yards at Kasumbalesa and Chililabombwe, Kapiri Mposhi, and Lusaka, where the main zone is already full. Kazungula's US$259.3m bridge brought Johannesburg-Lusaka down to about three days against four via Beitbridge, according to Freight News.

Mozambique and Namibia. Maputo handled a record 32.0 million tonnes in 2025. Walvis Bay's 2019 terminal, co-financed by the AfDB, raised capacity from 350,000 to 750,000 containers a year.

SUB-SAHARAN AFRICA: WEST AND CENTRAL

West Africa's ports compete for the Sahel; francophone markets lack a current rent benchmark

Côte d'Ivoire. Abidjan handled 46.6 million tonnes in 2025, up 16.1%. Mali-bound transit rose 76.4% to 1.47 million tonnes and Burkina Faso's 16.6% to 2.4 million, while Niger's transit has been diverted from the Cotonou route since the Malanville-Gaya crossing closed in July 2023, according to Agence Ecofin. The PK24 zone (1,000 ha, projected investment of FCFA 573.9bn, about US$986m at the 2025 average rate) and Ferkessédougou (732 ha) give the country more than 1,700 ha of state industrial and logistics land, delivered slowly.

Senegal. Dakar rose from 381st to 108th in the port index, the best Sub-Saharan rank in 2024. According to APA News, 65 to 70% of Mali's freight uses the Dakar-Bamako corridor. DP World's US$1.2bn Ndayane port, 50 km from Dakar near the airport, finished dredging in July 2026 ahead of an opening referred to as 2028, with an industrial zone beside it.

Togo and Benin. Lomé ran at about 89% of its rated capacity in 2025, and 92% of its transit was bound for Mali, Burkina Faso and Niger. Agence Ecofin reports that its integrated port and inland logistics cut Sahelian shippers' costs by 15 to 25%. In Benin, AGL is spending €75m (about US$81m at the 2024 average rate) to double Cotonou's quay to 1,100 m, the Glo-Djigbé zone covers 1,640 ha, and Benin's 2023 LPI score of 2.9 is above Ghana's 2.5 and Nigeria's 2.6.

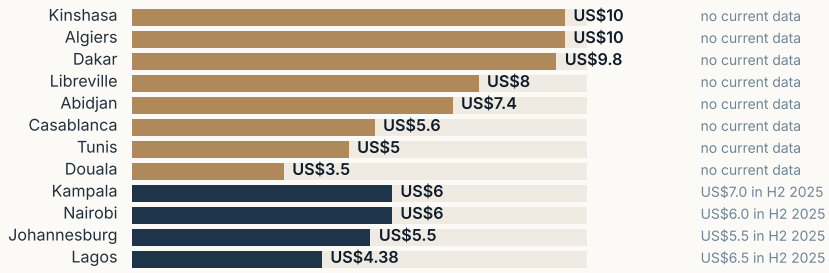
Cameroon, Congo and Gabon. Douala handled 12.9 million tonnes in 2024 (+6%) and Kribi 10.8 million; EcoMatin reports 555,398 TEU at Kribi in 2025. Kribi's 1,500 ha integrated industrial zone carries an estimated FCFA 550.3bn of investment (about US$907m at the 2023 average rate). Pointe-Noire's Congo Terminal grew from 200,000 TEU in 2009 to over a million, and Gabon's Nkok zone has more than 141 investors. The CEMAC central bank joined PAPSS in July 2026.

Nigeria and Ghana. Nigerian ports handled 129.3 million tonnes (+24.8%) and 2.1m TEU (+25.7%) in 2025. Lekki carried 40.6% of tonnage, but liquid bulk was 54.7% of the national total and transhipment 1.8%, so container-led warehouse demand still centres on Apapa and Tin Can. Lagos prime rents rose 48% since 2021 to US$6.5. Tema's MPS terminal expected about 2.3m TEU in 2025 and the port handles about 30% of Burkina Faso's imports.

EXHIBIT 21

Francophone cities had some of Africa's highest prime rents in 2021 and have no current benchmark

Prime industrial rent, H1 2021, US$ per m² a month; gold: cities dropped from Knight Frank's 2025 dashboards; current figure shown where tracked



None of the francophone cities in Knight Frank's 2021 survey appears in its 2025 dashboards. For investors in Abidjan, Dakar, Douala, Kinshasa, Casablanca or Algiers, the most recent Knight Frank prime rent dates from 2021, a gap that a French-language corridor rent survey could fill.

Source: Knight Frank, Africa Logistics & Industrial Review H2 2021 and Africa Industrial Market Dashboard H2 2025.

WHAT TO DO

Eighteen moves for the people who will own, finance, occupy and regulate corridor land

Each recommendation follows from evidence in this report and is addressed to the people who will act on it: occupiers, lenders, long-term investors, developers and the public bodies that control land, ports and borders.

CORPORATES CHOOSING WHERE TO HOLD REGIONAL STOCK

Boards of consumer goods, pharmaceutical, telecoms, industrial and logistics companies

1 **Check every lane against the schedule list.** On the latest published count, about a fifth of country pairs can use AfCFTA tariffs. Model tariff savings only for pairs where both sides have gazetted schedules, and use the US$5,000 self-declaration, 15% tolerance and cumulation for small regional consignments. Clothing and vehicles gain nothing until their rules of origin are agreed.

2 **Hold stock on the inland side of the slowest gate.** With 106.7 hours of dwell at Mombasa, a Nairobi distribution centre fed through Nairobi ICD may be closer in time to Kampala buyers than a coastal one. Apply the same test at Dar es Salaam and Kwala.

3 **Choose a depot by trains a day and customs on site.** Nairobi ICD runs at 69% of capacity and Naivasha at 1.8% on the same railway. Ask for frequency and clearance times before signing.

4 **Stress-test a route switch.** Mali-bound transit through Abidjan rose 76% in 2025, Niger's cargo left the Cotonou route when the Benin-Niger crossing closed in 2023, and transporters reported in 2021 that Kazungula saved Johannesburg-Lusaka trucks about a day over Beitbridge. Keep a second gateway in the plan.

DEVELOPMENT FINANCE INSTITUTIONS AND LENDERS

5 **Finance utilisation before capacity.** Rolling stock, signalling and on-site customs at existing depots are likely to move more cargo per dollar than new sheds. A depot at 1.8% of capacity needs trains.

6 **Report capital per tonne carried.** Make tonnes carried the covenant metric for rail and dry-port loans, with design capacity reported beside it, and push for published tariffs on new lines, starting with Lobito.

7 **Fund serviced land where queues exist.** Dongo Kundu and Lusaka South show demand waiting for serviced plots; a second phase at an oversubscribed zone carries less risk than a new zone.

PENSION FUNDS AND ASSET MANAGERS

8 **Underwrite flat dollar rents and price yield compression.** Prime rents are unchanged or lower in dollars since 2021 in most cities. The upside on corridor assets is a narrower inland yield premium: from 13% to 12% adds about 8% to value.

9 **Build the listed channel.** ALP Industrial REIT raised US$29.55m in 2026 against Equites' R28.7bn portfolio. Local pension capital can seed corridor logistics vehicles in Nairobi, Lagos, Abidjan and Casablanca.

10 **Use the Node Test to size positions.** Treat nodes scoring 8 or more as core, 6 to 7 as value-add with a named fix, and 5 or less as options on future evidence.

WHAT TO DO, CONTINUED

DEVELOPERS

11 **Buy within reach of the gate.** Container freight stations, empty depots and truck parks near Mombasa, Dar es Salaam, Durban and Lagos answer the hours lost at the port; bid for back-of-port leases such as Transnet's 25-year Bayhead sites.

12 **Build in towns where cargo breaks bulk.** Bonded yards, truck parks, clearing offices and workforce housing at Chirundu, Kasumbalesa and Nakonde-Tunduma; avoid pass-through posts.

13 **Phase mining-front land to rail finance.** Solwezi grew 232% before its rail link was financed; stage yards and housing there in step with the link.

14 **Fill the Grade A and cold-chain gap.** In Morocco 60% of warehouse sites are under 1,000 m^2 and cold storage covers about 4% of perishable volumes.

GOVERNMENTS, PORT AUTHORITIES, LAND AGENCIES AND INVESTMENT PROMOTION AGENCIES

15 **Publish dwell and depot throughput every month,** as the Northern Corridor already does, so that occupiers and investors can price the hours.

16 **Release serviced land beside working gates and depots** on long leases, and stop designating zones without trunk infrastructure: Kenya's gazetted zones had created 17% of the jobs projected by early 2024, according to Capital FM.

17 **Gazette tariff schedules.** Each new gazette notice widens the set of usable country pairs; for Côte d'Ivoire, Togo and Senegal it would also lift the case for their gateways.

18 **Design border posts with parking and bonded storage,** and put customs at the railhead. At Chirundu, time savings eroded as traffic grew.

THE RULE BEHIND THE EIGHTEEN

Follow the measured hours and tonnes. Treat announced capacity as a ceiling until traffic and tenants have used it.

TALK TO US

Heirstone Consulting helps corporates, investors and governments test corridor locations, value logistics and industrial land, and run the Node Test on specific sites with primary data.

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

TEU: twenty-foot equivalent unit. **Dwell:** average time containerised cargo spends in port, as published by NCTTCA. **ICD and dry port:** inland terminals where containers are cleared and stored away from the seaport. **OSBP:** one-stop border post, where both countries' agencies clear a vehicle once. **CPPI:** World Bank and S&P Global Container Port Performance Index, ranking ports on time ships spend in port. **Border compliance (DB2020):** time to obtain, prepare and submit documents and handle cargo at the main port or border for a standard import. **Gazetted schedule:** an AfCFTA tariff schedule given legal effect in domestic law. **Capacity in use:** latest throughput over design or rated capacity. **Effective speed:** straight-line distance between town centres divided by published transit time.

Method

Figures are as published by the sources listed, for the period stated. Where we derive a figure (capacity in use, capital per tonne, hours saved, growth rates, correlations, per-capita stock, country-pair shares) it is labelled Heirstone analysis and uses only the inputs shown. Local currencies are converted at World Bank annual average official rates for the year of the figure (2025: R17.89, 9.35 dirhams, KSh129.3, FCFA581.9, EGP49.2 and €0.885 to the US dollar; 2021: EGP15.6; 2023: €0.925; 2024: €0.924). The CFA franc is pegged to the euro. Where sources conflict we give the range or the more conservative figure and say so. Correlations across ten cities are indicative only. Several facts rest on a single secondary source; these are attributed in the text. Map positions are approximate town or port centres from GeoNames and Heirstone geocodes, and routes are schematic. The Node Test scores published evidence only; 0 can mean that no evidence was found.

Use of this report

This report is published for general information and uses public sources only. It is not investment, legal, tax or valuation advice and should not be relied on as such. Asking rents and announced investments are not transactions. Images are illustrative.

CHAPTER 1: THE TREATY

tralac, Status of AfCFTA ratification (4 Sep 2026) and 2025 Annual Conference two-pager; tralac rules of origin factsheet (Apr 2024); US International Trade Administration, Ghana AfCFTA updates (Apr 2024, Dec 2025); PAPSS (9 Jul 2026); Afreximbank African Trade Reports 2025 and 2026 via Agence Ecofin and Nairametrics; UNCTAD EDAR 2019; Brookings (2016, 2021); 57th ASEAN Economic Ministers' report via Vietnam MOIT; World Bank, The AfCFTA: Economic and Distributional Effects (2020).

CHAPTER 2: TIME AND PRICE

NCTTCA, 21st Transport Observatory Report (Aug 2026); Kenya News Agency citing KPA (Jan 2026); The Chanzo (Aug 2025); World Bank Doing Business 2020 economy profiles; TradeMark Africa OSBP portal; Central Corridor TTFA; JICA (2026); Freight News (2021, 2023, 2025); Maritimafrica citing DP World (Jul 2026); IRIS, Le corridor de Lobito (Nov 2025); World Bank, Teravaninthorn and Raballand (2009) and road freight study (2013); Djibouti Ports and Free Zones Authority via AllAfrica (Jan 2017).

CHAPTER 3: CAPACITY

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CHAPTER 4: PORTS AND BORDERS

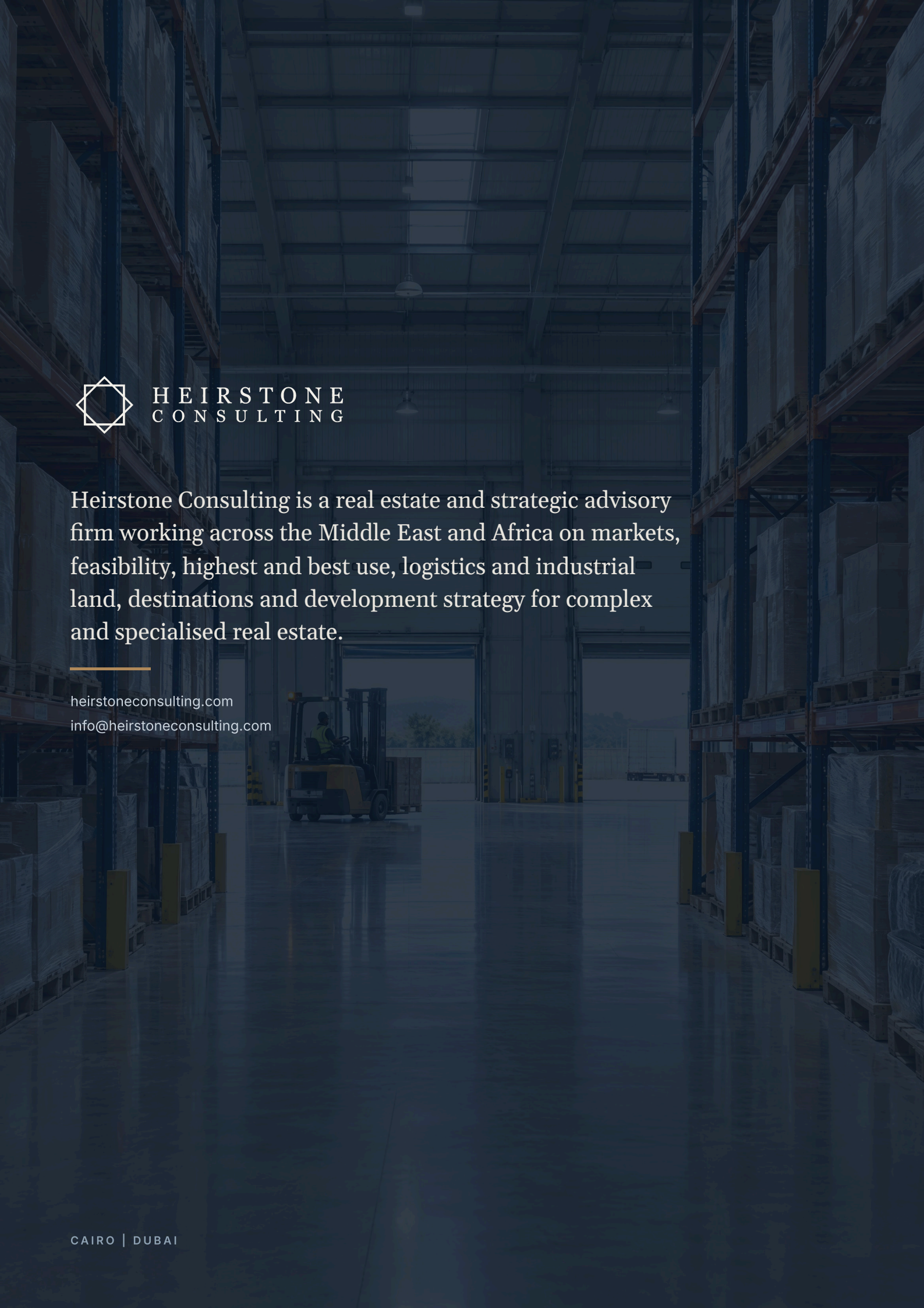
World Bank and S&P Global CPPI 2023; Agence Ecofin (Feb 2025, Sep 2025, May 2026); Seatrade Maritime (Jun 2024); Tanger Med Port Authority and Tanger Med Zones; Finances News Hebdo (Feb 2026); Lloyd's List via Freight News and Maritimafrica; Maritimafrica (2022-2026); City Population (Zambia and Zimbabwe censuses); tralac (Chirundu); conference presentation on OSBPs (D. Ndonga); Freight News (Apr 2019, Kasumbalesa); Construction Review Online (Aug 2021, Kazungula); AFP (Aug 2019, Walvis Bay); The Maritime Executive (Apr 2021, Dec 2025, Durban); The Citizen, Tanzania (Mar 2025) and Copperbelt Katanga Mining (Sep 2025, TAZARA); GeoNames (map positions).

CHAPTER 5: RENTS AND LAND

Knight Frank, Africa Logistics & Industrial Review H2 2021 and Africa Industrial Market Dashboards (Q1 2023, H1 and H2 2025); Business Daily Africa (2024, 2025); Cushman & Wakefield | Broll via IOL (Jul 2025) and Business Day (Jul 2026); JLL via BusinessTech (Aug 2024); IDC Zambia (Aug 2025); Capital FM (Jul 2026); Fraternité Matin and Team France Export (Feb 2026); Lagos Free Zone; Estate Intel; Finances News Hebdo citing AMDL (Aug 2025); World Bank WDI population and exchange rates.

CHAPTERS 6 AND 7

As above, plus Equites Property Fund; Serrari Group (2026); IPDC overview deck; ESL dry ports page; Fana (Jun 2024); Port Technology International (Jun 2021); EcoMatin (Nov 2025); GDIZ and Arise IIP; APA News (Nov 2025); Nigerian Ports Authority via Guardian Nigeria and Nairametrics (Mar 2026); Maritimafrica (Libya, Jan 2026); World Bank LPI 2023 and WDI.



HEIRSTONE
CONSULTING

Heirstone Consulting is a real estate and strategic advisory firm working across the Middle East and Africa on markets, feasibility, highest and best use, logistics and industrial land, destinations and development strategy for complex and specialised real estate.

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